

EXPERIMENT 22

AIM

Analyze simulated Windows authentication events, group failed logins by user and source address, and report threshold violations.

ALGORITHM

1. Store Windows authentication events.
2. Select failed-logon events using Event ID 4625.
3. Group failures by username and source IP.
4. Count failures in each group.
5. Compare counts with the threshold.
6. Report suspicious groups.
7. Stop.

CODE

```
from collections import defaultdict

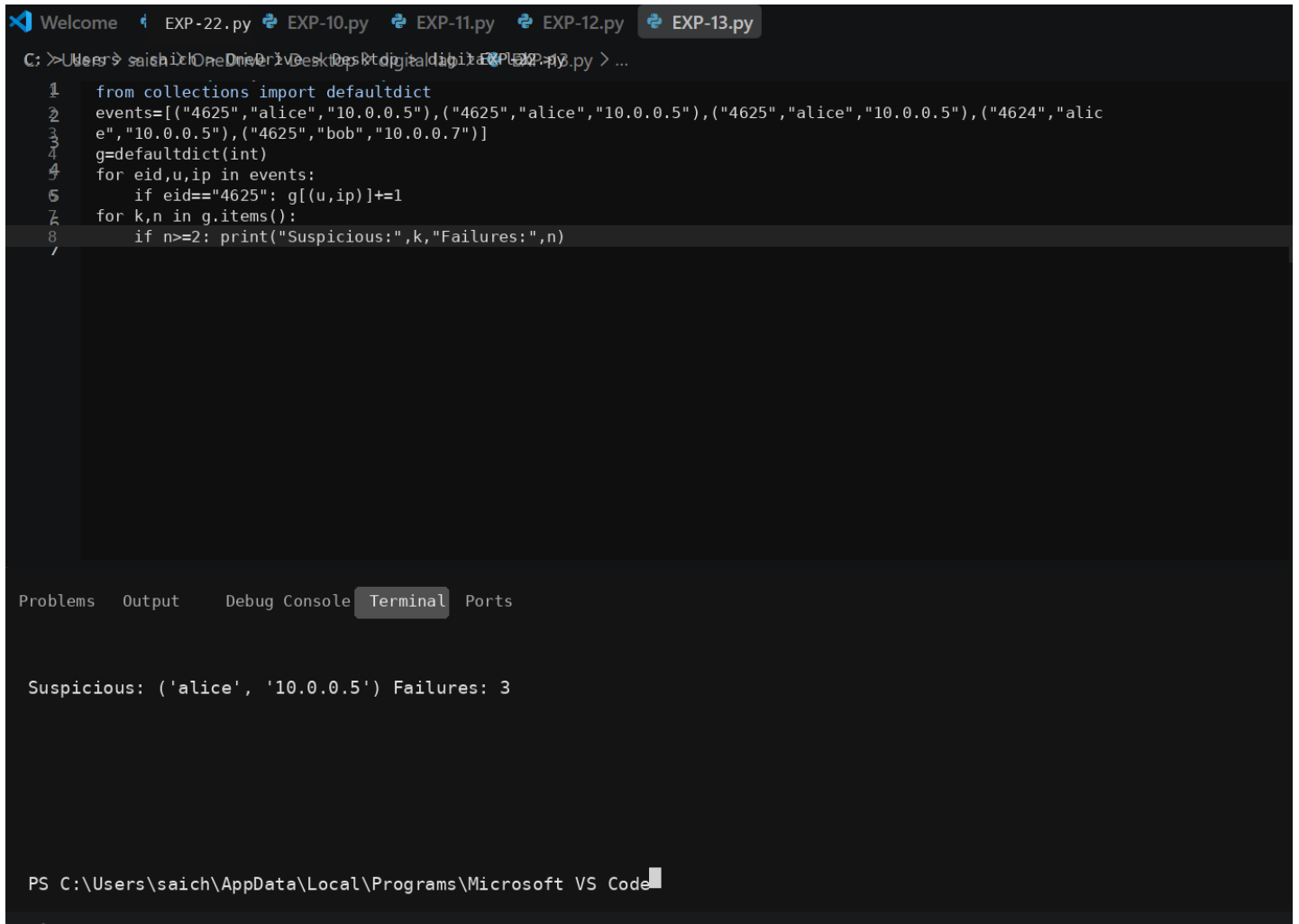
events = [
    ("4625", "alice", "10.0.0.5"),
    ("4625", "alice", "10.0.0.5"),
    ("4625", "alice", "10.0.0.5"),
    ("4624", "alice", "10.0.0.5"),
    ("4625", "bob", "10.0.0.7")
]

threshold = 2
failures = defaultdict(int)

for event_id, user, ip in events:
    if event_id == "4625":
        failures[(user, ip)] += 1

for (user, ip), count in failures.items():
    if count >= threshold:
        print("Suspicious:", (user, ip))
        print("Failures:", count)
```

OUTPUT



The screenshot shows a Visual Studio Code editor with a file explorer at the top displaying several Python files: EXP-22.py, EXP-10.py, EXP-11.py, EXP-12.py, and EXP-13.py. The main editor area shows the code for EXP-13.py. The code uses a defaultdict to track failed authentication events (eid, u, ip) and prints the results when the count for a specific user and IP combination reaches or exceeds 2.

```
1 from collections import defaultdict
2 events=[("4625","alice","10.0.0.5"),("4625","alice","10.0.0.5"),("4625","alice","10.0.0.5"),("4624","alic
3 e","10.0.0.5"),("4625","bob","10.0.0.7")]
4 g=defaultdict(int)
5 for eid,u,ip in events:
6     if eid=="4625": g[(u,ip)]+=1
7 for k,n in g.items():
8     if n>=2: print("Suspicious:",k,"Failures:",n)
9
```

Below the editor, the 'Terminal' tab is active, showing the output of the script:

```
Suspicious: ('alice', '10.0.0.5') Failures: 3
```

The status bar at the bottom indicates the file path: PS C:\Users\saich\AppData\Local\Programs\Microsoft VS Code

RESULT

The analyzer identifies repeated failed authentication events crossing the configured threshold.